Learning Objectives

1. Qualitatively determine whether an ionic salt solution is acidic, basic, or neutral by examining the effect on pH of the cation and anion.
2. For an ionic salt solution, calculate the pH and concentration of all species in solution.
3. Understand how a buffer resists pH change qualitatively.
4. Identify the composition of a buffer and relate to the principle of the common ion effect.
5. Calculate the pH of a buffer using the Henderson-Hasselbach equation
6. Calculate the pH changes due to addition of strong acid or strong base to a buffer system using the Henderson-Hasselbach equation

Ionization of Salts

- The conjugate acid/base of strong base/acid __do not__ react with water

$Cl^-(aq) + H_2O(l) \leftarrow HCl(aq) + OH^-(aq)$

base acid

$Na^+(aq) + H_2O(l) \leftarrow NaOH(aq) + H^+(aq)$

acid base

- The conjugate acid/base of a weak base/acid __do__ react with water

$CH_3COO^-(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$

base acid

$NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + H_3O^+(aq)$

acid base base

- Salts with anions (conjugate __base__ of a weak __acid__) makes __basic__ solution. OH^-
- Salts with cations (conjugate __acid__ of a weak __base__) makes __acidic__ solution. H^+

Your turn!

basic

Based on the table on the right, which salt solution will be acidic?

A. NaCl $\begin{cases} Na^+ \\ Cl^- \rightarrow \cancel{HCl} \end{cases}$ neutral

B. NH$_4$Cl $\begin{cases} NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+ \\ Cl^- \rightarrow \cancel{HCl} \end{cases}$ acidic

C. KNO$_2$ $\begin{cases} K^+ \\ NO_2^- \rightarrow HNO_2 + OH^- \end{cases}$ basic

$H_2O + NO_2$

D. KNO$_3$ $\begin{cases} K^+ \\ NO_3^- \rightarrow \cancel{HNO_3} \end{cases}$ neutral

Acid Strength	Acid	Base	Base Strength
Strong	HCl	Cl^-	Neutral
Strong	H_2SO_4	HSO_4^-	Neutral
Strong	HNO_3	NO_3^-	Neutral
Strong	H_3O^+	H_2O	Weak
Weak	HSO_4^-	SO_4^{2-}	Weak
Weak	H_2SO_3	HSO_3^-	Weak
Weak	H_3PO_4	$H_2PO_4^-$	Weak
Weak	HF	F^-	Weak
Weak	HNO_2		
Weak	$HC_2H_3O_2$	$C_2H_3O_2^-$	Weak
Weak	H_2CO_3	HCO_3^-	Weak
Weak	H_2S	HS^-	Weak
Weak	HSO_3^-	SO_3^{2-}	Weak
Weak	$H_2PO_4^-$	HPO_4^{2-}	Weak
Weak	HCN	CN^-	Weak
Weak	NH_4^+	NH_3	Weak
Weak	HCO_3^-	CO_3^{2-}	Weak
Weak	HPO_4^{2-}	PO_4^{3-}	Weak
Weak	H_2O	OH^-	Strong
Negligible	HS^-	S^{2-}	Strong
Negligible	OH^-	O^{2-}	Strong

Announcement 5/8

Assignments that are due soon

- Pre-Class Assignment Week 15-M

Office Hour

- Dr. Wang's: Fri 2-3 PM at KC 382

Salt with Anions as Weak Bases

- Anions that are conjugate base of a weak acid can produce OH^- in water

strong

A^- (aq) + H_2O(l) $\rightleftharpoons$ $HA(aq) + OH^-(aq)$ $\qquad K_b = \frac{K_w}{K_a} = \frac{10^{-14}}{K_a}$

base acid acid base

HA (aq) + H_2O(l) $\leftrightharpoons$ A^- (aq) + H_3O^+(l) $\qquad K_a \times K_b = K_w$

- Strong the acid (HA) is, weaker the conjugate base (A^-), less basic the salt solution will be.

NaOH (strong) HCHO2 (weak)

Example:
What is the pH of 0.100 M $NaCHO_2$ solution? K_a of $HCHO_2$ is 1.8 *10^{-4}.

1) Dissociation: $1\,NaCHO_2(aq) \rightarrow Na^+(aq) + 1\,CHO_2^-(aq)$

0.100

2) Ionization:

base acid

$CHO_2^-(aq) + H_2O(l) \rightleftharpoons HCHO_2(aq) + OH^-(aq)$

	CHO_2^-	H_2O	$HCHO_2$	OH^-
I	0.100		0	0
C	$-x$		$+x$	$+x$
E	$0.100-x$		x	x

$K_b = \frac{10^{-14}}{1.8\times10^{-4}} = 5.6\times10^{-11} = \frac{x^2}{0.100-x} \approx \frac{x^2}{0.100}$ $\qquad x = \sqrt{5.6\times10^{-11}\times0.100} = 2.4\times10^{-6}$

% ionization $= \frac{2.4\times10^{-6}}{0.1}\times100\% \ll 5\%$ $\qquad pOH = -\log(2.4\times10^{-6}) = 5.6$ $\qquad pH = 14 - 5.6 = 8.4$

Your turn!

Find the pH of 0.250 M $NaC_2H_3O_2$ solution. K_a of $HC_2H_3O_2$ is 1.8 *10^{-5}.

9.07

Your turn!

Find the pH of 0.250 M $NaC_2H_3O_2$ solution. K_a of $HC_2H_3O_2$ is 1.8 *10^{-5}.

$$NaC_2H_3O_2(aq) \rightarrow Na^+(aq) + C_2H_3O_2^-(aq)$$

base

$$C_2H_3O_2^-(aq) + H_2O(l) \rightleftharpoons HC_2H_3O_2(aq) + OH^-(aq) \qquad K_b = \frac{10^{-14}}{1.8\times10^{-5}} = 5.6\times10^{-10}$$

	$C_2H_3O_2^-$	H_2O	$HC_2H_3O_2$	OH^-
I	0.250	X	0	0
C	$-x$		$+x$	$+x$
E	$0.250-x$		x	x

$$(\_\_)^{\frac{1}{2}}$$

$$\frac{x^2}{0.250-x} \approx \frac{x^2}{0.250} = 5.6\times10^{-10}$$

$$x = \sqrt{(5.6\times10^{-10}\times0.250)} = 1.2\times10^{-5} = [OH^-]$$

$$pOH = -\log(1.2\times10^{-5}) = 4.93$$

$$pH = 14 - 4.93 = 9.07 > 7$$

9.07

Salt Cations as Weak Acid

- Cations that are conjugate ~~base~~ acid of a weak ~~acid~~ base can produce $\underline{H_3O^+}$ in water

acid base base acid

$\rightarrow BH^+ (aq) + H_2O(l) \rightleftharpoons B(aq) + H_3O^+(aq)$ $\qquad K_a = \frac{K_w}{K_b} = \frac{10^{-14}}{K_b}$

base acid

$\rightarrow B (aq) + H_2O(l) \leftrightharpoons BH^+ (aq) + \cancel{H_3O^+(l)}\ OH^-(aq)$ $\qquad K_b$

- Strong the base is, __weaker__ the conjugate acid, __less__ acidic the salt solution will be.

B BH+

→ smallest → largest

Table 17.3 K_b and pK_b Values for Weak Molecular Bases at 25 °C

Name of Base	Formula	K_b	$pK_b = -\log(K_b)$
Butylamine	$C_4H_9NH_2$	5.9×10^{-4}	3.23
Methylamine	CH_3NH_2	4.4×10^{-4}	3.36
Ammonia	NH_3	1.8×10^{-5}	4.74

Which solution is the most acidic solution?

$Cl^- \leftarrow HCl$
neutral

A. $C_4H_9NH_3^+ | Cl^-$ $\qquad C_4H_9NH_2$

B. $CH_3NH_3^+ | Cl^-$ $\qquad CH_3NH_2$

C. $NH_4^+ | Cl$ $\qquad NH_3$

acid strongest $\qquad$ base weakest

Your turn!

Determine the pH of a 0.15 M solution of NH_4Cl. K_b of NH_3 is 1.8 *10^{-5}.

$$NH_4Cl(aq) \rightarrow NH_4^+(aq) + Cl^-(aq)$$

$$NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + H_3O^+(aq)$$

Your turn!

Determine the pH of a 0.15 M solution of NH_4Cl. K_b of NH_3 is 1.8 *10^{-5}.

$$NH_4Cl(aq) \rightarrow NH_4^+(aq) + Cl^-(aq)$$

$$NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + \boxed{H_3O^+(aq)}$$

	NH_4^+	H_2O	NH_3	H_3O^+
I	0.15	X	0	0
C	$-x$		$+x$	$+x$
E	$0.15-x$		x	x

$$K_a = \frac{10^{-14}}{1.8\times10^{-5}} = 5.6\times10^{-10}$$

$$\frac{x^2}{0.15-x} \approx \frac{x^2}{0.15} = 5.6\times10^{-10} \qquad x^2 = 5.6\times10^{-10}\times0.15$$

$$x = \sqrt{5.6\times10^{-10}\times0.15} = 9.1\times10^{-6} = [H_3O^+]$$

$$pH = -\log(9.1\times10^{-6}) = 5.04$$

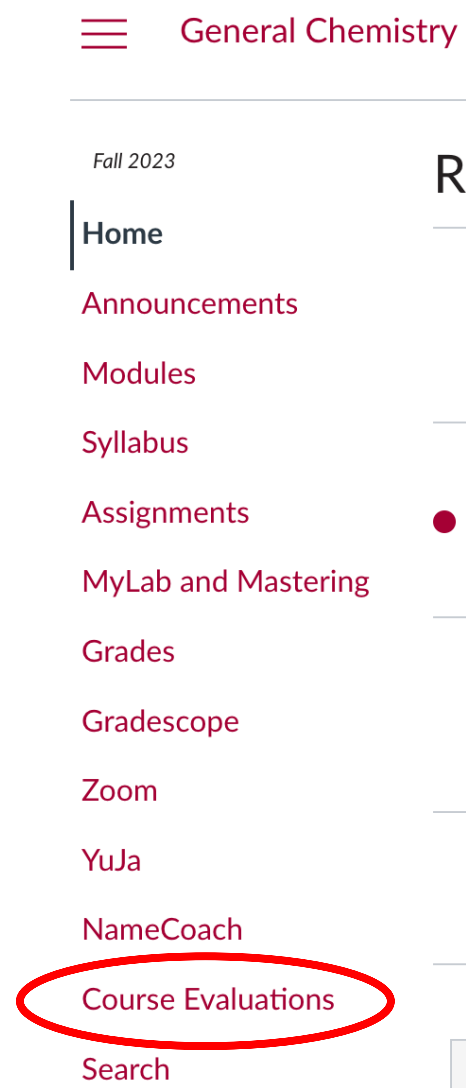
5.04

Course Evaluation

Everyone get one bonus points for your final exam if the class response rate is 85% or higher.

Please fill Course Evaluation. Your feedback are valued!

- Anonymous
- What part you like on the class (helped you study)
- Your suggestions on making the class better



Announcements 5/11

Final Exam at 8:00 AM on Tue 5/19:

- 2 hour 30 minutes long
- Calculators (any kind), pens/pencils/highlighters, and erasers are allowed.
- Learning objective uploaded

Assignments that are due soon

- Pre-Class Assignment Week 15-W

Office Hour

- Dr. Wang’s: Tue 10-11 AM at KC 230
- Paige Kwan’s: Mon 11:30 AM -12:30 PM at the TLC (Demille 160P)
- Saige Douglass’s: Tue 12:00-1:00 PM at the TLC (Demille 160P)

Common Ion Effect

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

strong acid pH=1

$$K_a = \frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]} = 1.8 * 10^{-5}. \ll 1$$

0.10 M CH_3COOH solution, pH = ?	0.10 M CH_3COOH + 0.10 M CH_3COONa, pH = ?

0.10 M CH_3COOH solution, pH = ?

	$CH_3COOH(aq)$	$+ H_2O(l) \rightleftharpoons$	$CH_3COO^-(aq)$	$+ H_3O^+(aq)$
I	0.10	X	0	0
C	$-x$		$+x$	$+x$
E	$0.1-x$		x	x

$$\frac{x^2}{0.1-x} \approx \frac{x^2}{0.1} = 1.8\times10^{-5} \quad x = 1.34\times10^{-3}$$

$$\%\text{ionization} = \frac{1.34\times10^{-3}}{0.1}\times100\% = 1.34\% < 5\%$$

$$pH = -\log(1.34\times10^{-3}) = \boxed{2.87} < 7$$

0.10 M CH_3COOH + 0.10 M CH_3COONa, pH = ?

	$CH_3COOH(aq)$	$+ H_2O(l) \rightleftharpoons$	$CH_3COO^-(aq)$	$+ H_3O^+(aq)$
I	0.10	X	0.10	0
C	$-x$		$+x$	$+x$
E	$0.10-x$		$0.10+x$	x

$$Q(I) = \frac{0.10\times0}{0.10} = 0 < K$$

$$\frac{(0.10+x)\times x}{0.10-x} \approx \frac{\cancel{0.10}\,x}{\cancel{0.10}} = 1.8\times10^{-5}$$

$$\%\text{ionization} = \frac{1.8\times10^{-5}}{0.10}\times100\% = 1.8\times10^{-2}\%$$

$$pH = -\log(1.8\times10^{-5}) = \boxed{4.74}$$

Adding CH_3COO^-, a common ion, the ionization of CH_3COOH has been suppressed.

Common Ion Effect

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

$$K_a = \frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]} = 1.8 * 10^{-5}.$$

0.10 M CH_3COOH solution, pH = ?

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

	CH_3COOH	H_2O	CH_3COO^-	H_3O^+
I	0.10	X	0	0
C	$-x$		$+x$	$+x$
E	$0.1-x$		x	x

$$\frac{x^2}{0.1-x} \approx \frac{x^2}{0.1} = 1.8\times10^{-5}$$

$$x = \sqrt{1.8\times10^{-5} \times 0.1} = 1.34\times10^{-3}$$

$$pH = -\log(1.34\times10^{-3}) = 2.87$$

0.10 M CH_3COOH + 0.10 M CH_3COONa, pH = ?

$$CH_3COONa(aq) \rightarrow CH_3COO^-(aq) + Na^+(aq)$$

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

	CH_3COOH	H_2O	CH_3COO^-	H_3O^+
I	0.1	X	0.1	0
C	$-x$		$+x$	$+x$
E	$0.1-x$		$0.1+x$	x

$$\frac{(0.1+x)x}{0.1-x} \approx \frac{\cancel{0.1}\cdot x}{\cancel{0.1}} = 1.8\times10^{-5}$$

$$[H_3O^+] = x = 1.8\times10^{-5}$$

$$pH = -\log(1.8\times10^{-5}) = 4.74$$

Adding CH_3COO^-, a common ion, the ionization of CH_3COOH has been suppressed.

Henderson-Hasselbach Equation

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

$$\Rightarrow K_a = \frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]}$$

$$\boxed{-\log(Ka)} = -\log\left(\frac{[\text{Conjugate base}][H^+]}{[\text{weak acid}]}\right)$$

$$pKa = -\log\left(\frac{[\text{conjugate base}]}{[\text{weak acid}]}\right) \boxed{-\log[H^+]}$$

$$pKa = \boxed{-\log\left(\frac{[\text{conjugate base}]}{[\text{weak acid}]}\right)} + pH$$

$$pH = pK_a + \log\left(\frac{[\text{conjugate base}]}{[\text{weak acid}]}\right) = pK_a + \log\left(\frac{[A^-]}{[HA]}\right)$$

$$pH = pKa + \log\left(\frac{[A^-]}{[HA]}\right)$$

Henderson-Hasselbach Equation

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

$$K_a = \frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]}$$

$$-\log K_a = -\log\left(\frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]}\right)$$

$$pK_a = -\log\left(\frac{[CH_3COO^-]}{[CH_3COOH]}\right) - \log([H_3O^+])$$

$$pK_a = -\log\left(\frac{[CH_3COO^-]}{[CH_3COOH]}\right) + pH$$

$$pH = pK_a + log\left(\frac{[conjugate\ base]}{[weak\ acid]}\right) = pK_a + log\left(\frac{[A^-]}{[HA]}\right)$$

Your turn! $\Rightarrow pH = pK_a + log(\frac{[A^-]}{[HA]})$

Find the pH for a solution mixed with 0.10 M CH_3COOH and 0.20 M CH_3COONa.
The pK_a of CH_3COOH is 4.74.

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

$$pH = 4.74 + \log\left(\frac{0.2}{0.1}\right) = 5.04$$

5.04

Buffer: Solutions That Resist pH Change

- A buffer solution can resist changes in pH upon the addition of small amounts of acid/base
- To make a buffer, it **MUST** contain :
  1. A weak acid/base (K_a/K_b << 1)
  2. A salt that contains the conjugate base/acid

Which of the following combination of solutions can be used to make buffer?

a) KF/HF ($K_a(HF) = 6.6*10^{-4}$) ✓

b) KBr/HBr ($K_a(HBr) = 1.0*10^{9}$) >>1 strong acid

$NaHCO_3$

c) Na_2CO_3/H_2CO_3 ($K_a(H_2CO_3) = 4.3*10^{-7}$) <<1

CO_3^{2-} HCO_3^-

✓ $Na_2CO_3/NaHCO_3$

✓ $NaHCO_3/H_2CO_3$

Buffer: Solutions That Resist pH Change

- A buffer solution cane resist changes in pH upon the addition of small amounts of acid/base
- To make a buffer, it **MUST** contain :
  1. A weak acid/base (K_a/K_b<<1)
  2. A salt that contains the conjugate base/acid

$\rightarrow CH_3COO^- + Na^+$

$pH = \boxed{4.74}$

The original solution is a 1 L of the solution of 0.1 M CH_3COONa and 0.1 M CH_3COOH. The pK_a of CH_3COOH is 4.74. What is the pH of the solution after adding 0.03 mol NaOH. $\rightarrow Na^+ + OH^-$

LR

$$OH^-(aq) + CH_3COOH(aq) \rightarrow H_2O(l) + CH_3COO^-(aq)$$

	OH^-	CH_3COOH	H_2O	CH_3COO^-
I	0.03	0.1	X	0.1
C	-0.03	-0.03		+0.03
F	0	0.07		0.13

$$pH = 4.74 + \log\left(\frac{0.13}{0.07}\right) = \boxed{5.01}$$

In water. $pOH = -\log(0.03) = 1.5$ $\quad pH = 12.5$

Your turn!

$$pH = pK_a + log(\frac{[A^-]}{[HA]})$$

base.

The original solution is a 1 L of the solution of 0.1 M CH_3COONa and 0.1 M CH_3COOH. The pK_a of CH_3COOH is 4.74. What is the pH of the solution after adding 0.03 mol HCl. (Hint: HCl will completely react with some of CH_3COONa)

acid

$\hookrightarrow CH_3COO^- + Na^+$ $\hookrightarrow H^+ + Cl^-$

$$H^+(aq) + CH_3COO^-(aq) \rightarrow CH_3COOH(aq)$$

$$HCl(aq) + CH_3COONa(aq) \rightarrow CH_3COOH(aq) + NaCl(aq)$$

4.47

Your turn!

$$pH = pK_a + log(\frac{[A^-]}{[HA]})$$

The original solution is a 1 L of the solution of 0.1 M CH_3COONa and 0.1 M CH_3COOH. The pK_a of CH_3COOH is 4.74. What is the pH of the solution after adding 0.03 mol HCl. (Hint: HCl will completely react with some of CH_3COONa)

$$HCl(aq) + CH_3COONa(aq) \rightarrow CH_3COOH(aq) + NaCl(aq)$$

	HCl	CH_3COONa	CH_3COOH	NaCl
I	0.03	0.1	0.1	0
C	−0.03	−0.03	+0.03	+0.03
F	0	0.07	0.13	0.03

0.1M Before pH = 4.74

$$pH = 4.74 + \log\left(\frac{0.07}{0.13}\right) = \boxed{4.47}$$

1M Before, 4.74 After 4.71

4.47

Announcement 5/13

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

SI Session

Saige Douglass – Wed 5:30 – 7:30 PM at Hashinger 412

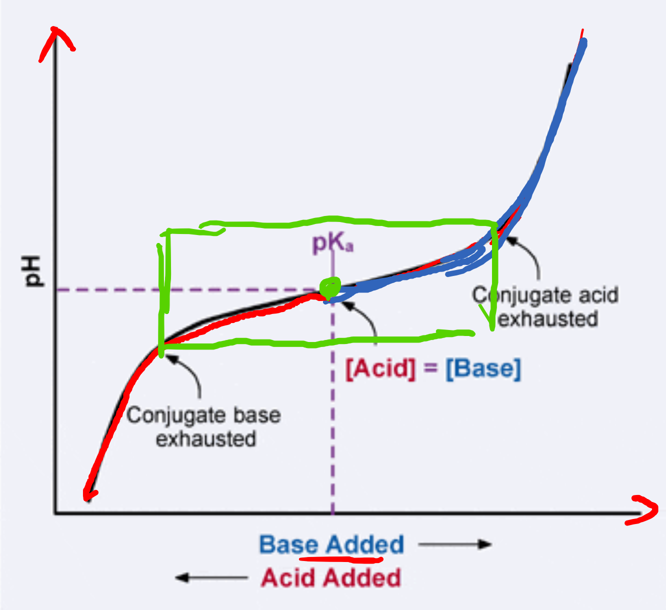
Paige Kwan - Thu 5 - 7 PM at KC134

Buffer: Solutions That Resist pH Change

- To make the best buffer: $\frac{[A^-]}{[HA]}$ = 1

$$pH = pK_a + \log\frac{[A^-]}{[HA]}$$

- pH = pKa



- The best choice to make a buffer with a pH at:

1) 7.4 (Blood)

$H_2PO_4^- / HPO_4^{2-}$

2) 2.5 (stomach)

$H_3PO_4 / H_2PO_4^-$

Acid (name)	Conjugate Base (name)	pK_a	K_a
HCOOH (formic acid)	$HCOO^-$ (formate ion)	3.8	1.78×10^{-4}
CH_3COOH (acetic acid)	CH_3COO^- (acetate ion)	4.8	1.74×10^{-5}
H_3PO_4 (phosphoric acid)	$H_2PO_4^-$ (dihydrogen phosphate ion)	2.1	7.24×10^{-3}
$H_2PO_4^-$ (dihydrogen phosphate ion)	HPO_4^{2-} (hydrogen phosphate ion)	6.9	1.39×10^{-7}
HPO_4^{2-} (hydrogen phosphate ion)	PO_4^{3-} (phosphate ion)	12.4	3.98×10^{-4}
H_2CO_3 (carbonic acid)	HCO_3^- (bicarbonate ion)	6.3	5.1×10^{-7}
HCO_3^- (bicarbonate ion)	CO_3^{2-} (carbonate ion)	10.3	5.62×10^{-11}
NH_4^+ (ammonium)	NH_3 (ammonia)	9.3	5.62×10^{-10}

Buffer Capacity

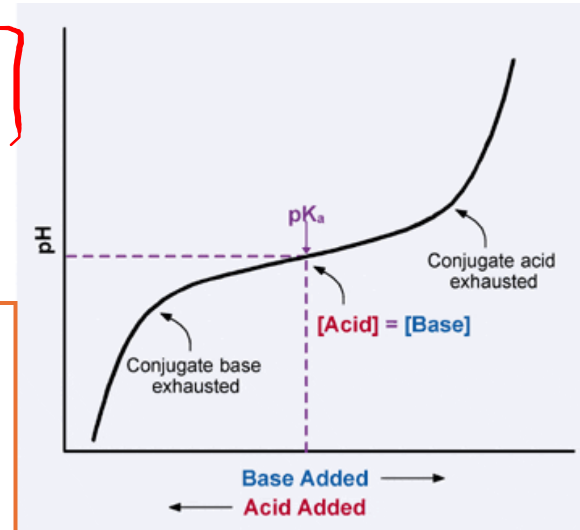
- A higher concentration of buffer, the $\frac{[A^-]}{[HA]}$ change will be smaller, the buffer capacity is higher.

$H^+ + CH_3COO^- \rightarrow CH_3COOH$

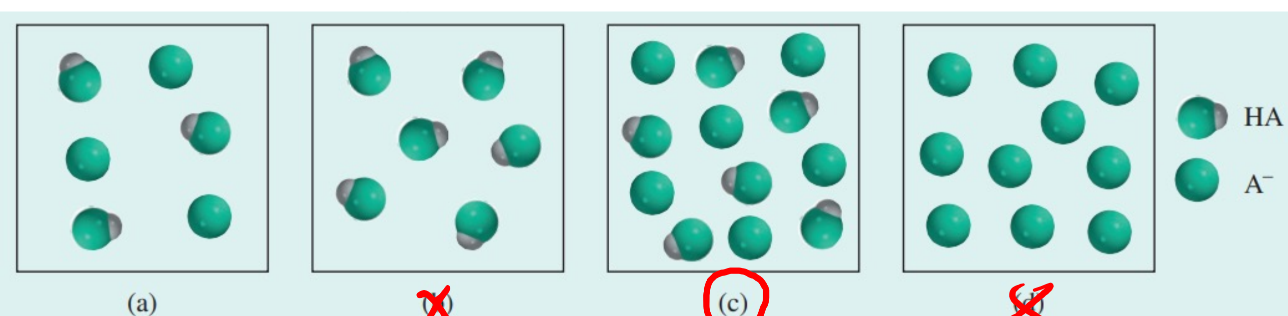
Calculate the pH change of adding 0.03 mol of HCl into the buffer of 1 M CH_3COONa and 1 M CH_3COOH. The pK_a of CH_3COOH is 4.74.

$$pH = pK_a + log\frac{[A^-]}{[HA]} = 4.74 + \log\left(\frac{1-0.03}{1+0.03}\right) = \boxed{4.71}$$

Before $\boxed{pH = 4.74}$



The diagrams below represent solutions containing a weak acid HA and/or its sodium salt NaA. Which solutions can act as a buffer? Which solution has the greatest buffer capacity? The Na^+ ions and water molecules are omitted for clarity.



Your turn!

$$pH = pK_a + log(\frac{[A^-]}{[HA]})$$

To make a buffer solution that has a pH of 5.32, how much, in grams, of CH_3COONa (mm = 82.03 g/mol) needs to add into 1L of 0.1 M CH_3COOH. The pK_a of CH_3COOH is 4.74.

$$5.32 = 4.74 + \log\left(\frac{[CH_3COO^-]}{0.1\,M}\right)$$

$$\log_{10}\left(\frac{[CH_3COO^-]}{0.1\,M}\right) = \underline{0.58}$$

$$\frac{[CH_3COO^-]}{0.1\,M} = 10^{0.58}$$

$$[CH_3COO^-] = 3.80 \times 0.1 = 0.38M$$

$$\frac{0.38\,mol}{1L} \times 1L \times \frac{82.03g}{1mol} = \boxed{31.17\,g}$$

31.17 g